DSCI 560 Final Project Proposal
Freelancer-Client AI Communication Platform

Team Pylovers

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1. Team Organization

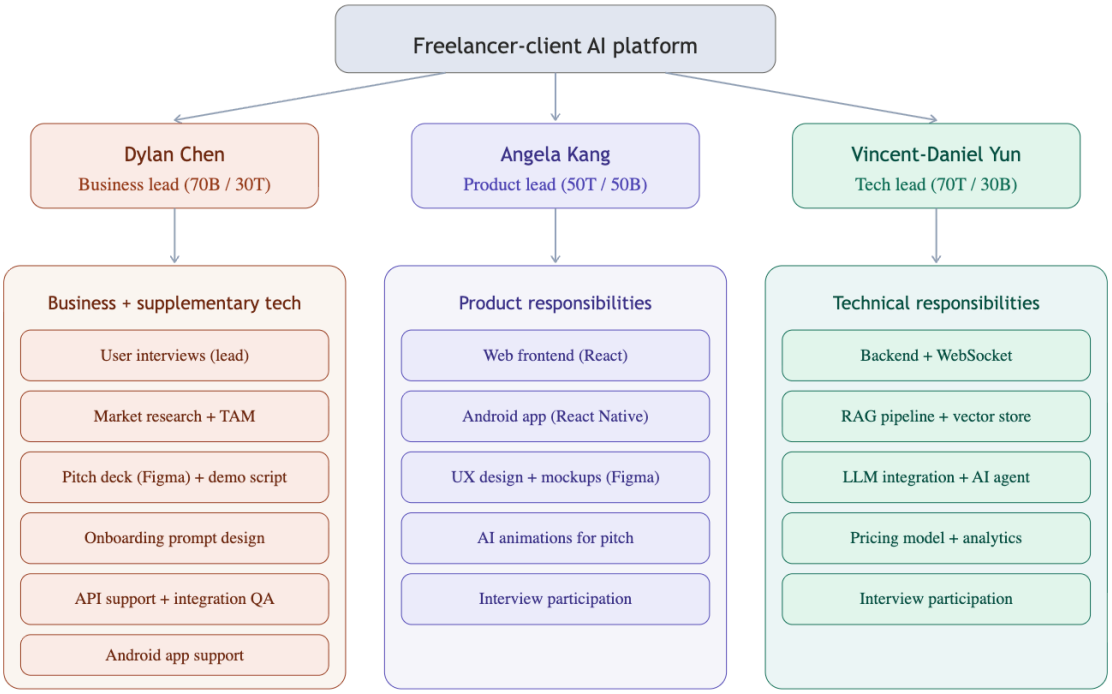

Diagram 1: Team Organization

Role	Contributor	Split	Description
Business Lead	Dylan Chen	70% Business / 30% Technical	Market strategy, user interviews, pitch deck (Figma), demo script (lead), onboarding prompt engineering, competitive positioning, supplementary technical across all tiers
Tech Lead	Vincent-Daniel Yun	70% Technical / 30% Business	Backend API, WebSocket server, RAG pipeline, LLM integration, AI scope guardian (lead), database schema, pricing model, interview participation
Product Lead	Angela Kang	50% Technical / 50% Business	Web frontend, Android app, UX design, high-fidelity mockups (lead), interview participation, acquisition strategy

All members contribute to both business and technical aspects with cross-functional overlap. All members will participate in user interviews, demo production, regular meetings, documentation, and code review.

2. Technical Architecture

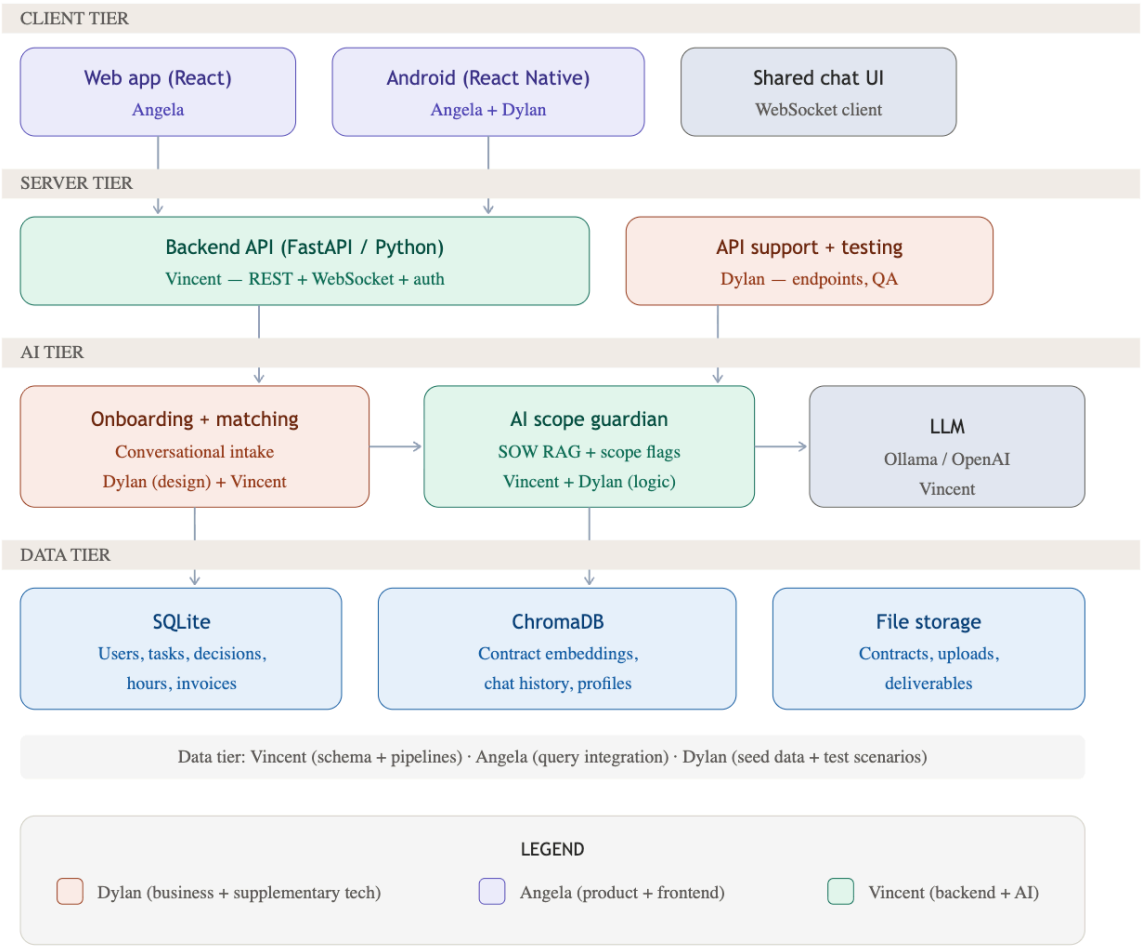

Diagram 2: Technical Architecture

The platform implementation is organized into four layers:

Client Layer

- React Web App (Angela) - responsive chat UI, project dashboard, onboarding flows
- React Native Android App (Angela) - shared component library with web, push notifications
- Shared WebSocket Client - real-time chat across both platforms

Server Layer

- FastAPI/ Python Backend (Daniel) - REST API endpoints, WebSocket hub, authentication, session management
- Routes chat messages, manages project state, handles file uploads, coordinates AI agent calls

AI Layer

- Onboarding + Matching Agent (Dylan: prompts and conversation design / Daniel: implementation) - LLM-assisted conversational user intake for freelancers and clients, embeddings-based matching
- AI Scope Guardian (Daniel) - monitors project chat. Cross-references client requests against contract/SOW via RAG, generates private scope creep flags to freelancers, logs decisions and milestones
- LLM Layer (Daniel) - Ollama for local development or OpenAI API, function calling for structured actions

Data Layer

- SQLite (Daniel: schema / Angela: query integration) - users, projects, tasks, decisions, time entries, invoices, match history
- Vector DB (Daniel) - vector store for contract/SOW embeddings, chat history embeddings, freelancer/client profile embeddings for matching
- File storage - uploaded contracts, deliverables, images

Technical Component Ownership

Component	**Primary Owner**	**Support**
WebSocket Chat Backend	Daniel	Angela
REST API + Auth	Daniel	Angela
RAG Pipeline (Contract -> Embeddings -> Vector DB)	Daniel	Dylan
AI Scope Guardian	Daniel	
LLM Integration (e.g., Ollama function calling)	Daniel	
Matching Engine (Embedding Similarity)	Dylan	Daniel
Web Frontend (React)	Angela	
Android App	Angela	
UI/UX Design + High-Fidelity Mockups	Angela	Dylan
File Upload + Processing	Angela	Daniel
Database Schema	Daniel	Angela
Data Collection / Seed Data Pipeline	Dylan	All

3. Business Architecture

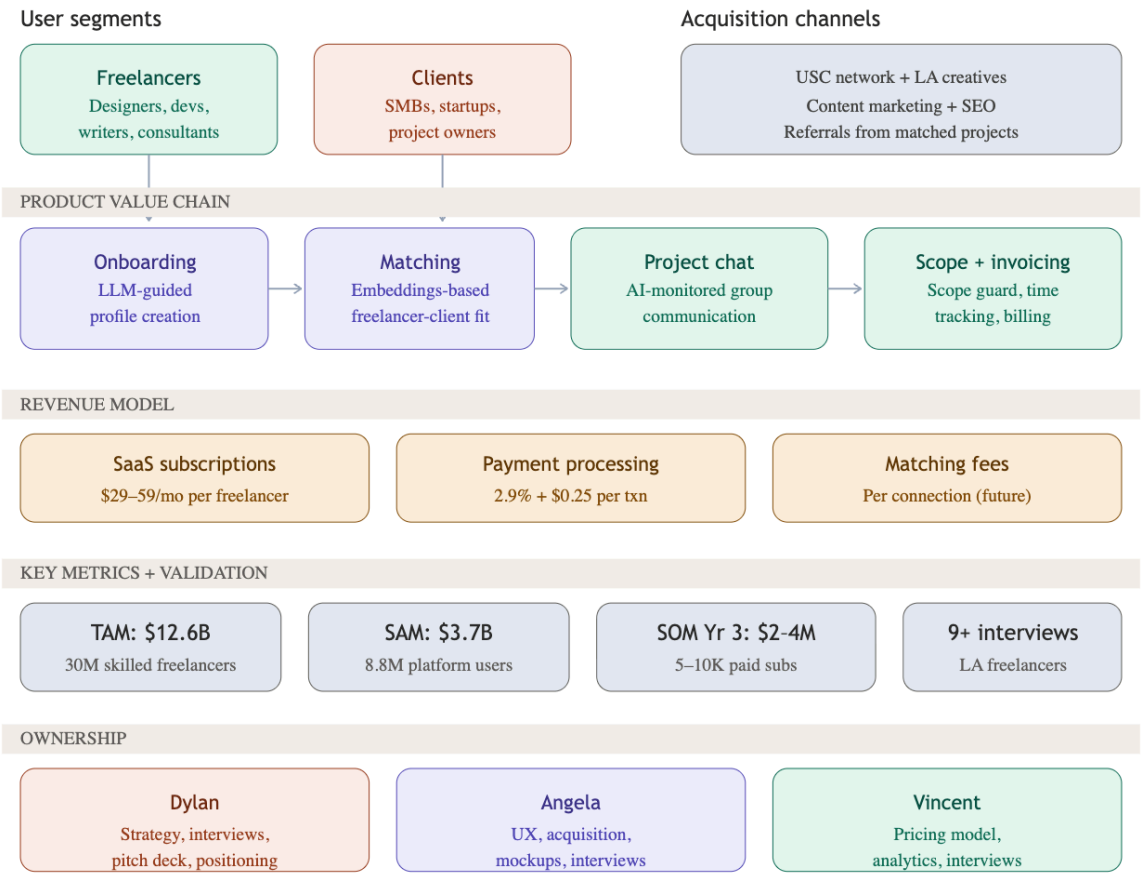

Diagram 3: Business Architecture

User Segments

- Freelancers - designers, developers, writers, consultants (primary paying customer)
- Clients - startups, project owners, SMBs

Product Value Chain

- Onboarding - LLM-assisted conversational profile creation (in replacement of forms)
- Matching - embeddings-based freelancer-client matching with AI explanations
- Project Chat - AI-monitored group communication (freelancer + client(s) and stakeholders + AI agent)
- Scope + Invoicing - scope guardian, decision logging, time tracking, invoice generation, meeting summaries

Revenue Model

- SaaS Subscriptions - freemium monthly model (Free / Pro / Business tiers)

- Payment Processing - percentage fee per transaction (i.e., Stripe)
- Matching Fees - per-connection fee (tentatively)

Acquisition Channels

- Startups / Accelerators and Design / Engineering communities
- LA creative freelancer community (initial)
- Content marketing and SEO targeting freelancer pain points
- Referral loops from successful project matches

Business Component Ownership

Component	**Primary Owner**	**Support**
Market Research and Competitive Analysis	Dylan	All
User Interviews	Dylan (lead)	All
Pricing and Monetization Strategy	Dylan	Angela
Investor Pitch Deck (Week 5 Target)	Dylan	All
User Persona Development	Angela	Dylan
UX Research from Interviews	Angela	Dylan
Competitive UX Benchmarking (tentatively)	Angela	
Competitor Technical Analysis	Daniel	All
Demo Video Script and Recording	All	
Cold Start Strategy	Dylan	All

4. Weekly Development Schedule

The schedule front-loads technical building in Weeks 1–3 while user interviews run in parallel. Weeks 4–5 shift toward refinement, pitch preparation, and integration of interview findings.

Week 0 (3/22 - 3/28):

- Initial Project Ideas Proposals
- Beginning User Interviews (1 product/ui/ux designer, 2 software engineers, 1 video editor)
- Finishing and submitting Final Project Proposal DRAFT 2

Week 1 (3/29 - 4/4):

- Conduct user interviews 1–3 (Dylan lead, Angela support)
- Build WebSocket chat server and REST API skeleton with JWT auth (Daniel)
- Implement SQLite database schema (Daniel)
- Begin RAG pipeline: contract extraction and chunking via pdfplumber (Daniel)
- Build onboarding chat flow prototype and begin matching engine (Dylan)
- Build chat UI in React and connect to WebSocket backend (Angela)
- Begin Android app chat screen (Angela)
- Milestone: Basic chat works end-to-end

Week 2 (4/5 - 4/11):

- Conduct user interviews 4–6 (Dylan lead, Angela support)
- Complete RAG pipeline: contract → chunks → embeddings → ChromaDB (Daniel)
- Implement AI Scope Guardian: message → vector query → LLM → scope flag (Daniel)
- Integrate AI agent into chat as private notes to freelancer (Daniel)
- Complete matching engine: embedding similarity scoring and ranked results (Dylan)
- Build project dashboard UI and file upload integration (Angela)
- Start invoice and time-tracking UI components (Angela)
- Milestone: AI scope monitoring works end-to-end

Week 3 (4/12 - 4/18):

- Conduct user interviews 7–9, compile findings report (Dylan lead)
- Integrate onboarding and matching flow into main app; seed database (Dylan)
- Decision logger: tag agreements in chat with timestamps (Daniel)
- Refine scope monitoring accuracy; API hardening and error handling (Daniel)
- Polish web UI; complete Android app; invoice generation flow (Angela)
- Milestone: Full flow: onboard → match → chat → scope flag → invoice

Week 4 (4/19 - 4/25):

- Finalize business case, pricing, and pitch deck outline (Dylan)
- Draft demo video script (Dylan)
- Create high-fidelity mockups for all major screens (Angela)
- Begin AI animation assets for pitch (Angela, REQUIRED)

- Performance tuning, bug fixes, scope monitor edge cases (Daniel)
- UI polish pass and cross-platform testing (Angela)
- Milestone: Feature-complete and stable. Hi-fi mockups ready.

Week 5 (4/26 - 4/30):

- PRODUCT PITCH
- Finalize pitch deck and lead rehearsals (Dylan)
- Prepare technical demo walkthrough; ensure system stability (Daniel)
- Complete AI animation assets; rehearse UX walkthrough (Angela)
- Deliverable: Pitch presentation, high-fidelity mockups, AI animations

Week 6 (5/1 - 5/9):

- Incorporate pitch feedback across product and business narrative
- Finalize and record demo video (all members)
- Final integration testing and realistic demo data seeding (Daniel)
- Final cross-platform testing and UI polish (Angela)

Week 6.5 (5/10 - 5/12):

- Finalize product and working demonstration video
- Final system check and demo environment stability (Daniel)-
- Final video editing and visual polish (Angela)
- Submit all final deliverables (Dylan)

Deliverable for Pitch

- Emotional connection and financial viability
- High-fidelity mockups (no live demo required)
- AI Animations REQUIRED
- Convince Investors

5. Key Milestones Summary

The following milestones define our critical path from initial proposal through final delivery. Technical gates ensure core infrastructure is validated before AI integration begins. Integration and polish gates confirm the product is stable and pitch-ready before the final submission.

Week	**Milestone**	**Gate**
0	Final Project Proposal Draft 2 submitted	Completed
1	Basic chat works end-to-end (WebSocket on web)	Technical
2	AI Scope Guardian functional (contract upload → scope flag in chat)	Technical
3	Full flow: onboard → match → chat → scope flag → invoice	Integration
4	Feature-complete and stable. High-fidelity mockups ready for pitch	Polish
5	Product Pitch	Presentation
6	Demo video recorded	Content
6.5	Final product and demonstration video submitted	Ship Product

6. Interview Schedule

Interviews are front-loaded in Weeks 0–3, with follow-ups as needed through Week 4. We conduct separate interview tracks for freelancers and clients to capture both sides of the scope creep problem.

Sample freelancer questions include: how they track client requests versus original agreements, how they handle out-of-scope asks, and what tools or systems they wish existed. Sample client questions include: how they stay informed on project progress, whether they've ever disagreed with a freelancer on what was included in the original agreement, and what would make working with freelancers easier.

Week	**Interviews**	**Lead**	**Goal**
0	Freelancers and clients across roles: product/UI/UX designers, software engineers, video editors	Dylan (lead), Angela (support)	Validate core pain point. Confirm scope creep and communication overhead are top concerns.
1	Freelancers (5+ concurrent clients) and clients who have managed multiple freelance engagements	Dylan (lead), Angela (support)	Deepen understanding of scope creep incidents, current tooling gaps, and unmet needs. Identify recurring frustrations in the client-freelancer communication cycle.
2	Mix of freelancers and clients, prioritizing those with contract or invoicing experience	Dylan (lead), Angela (support)	Validate pricing sensitivity and willingness to pay. Prioritize features based on stated pain points. Probe on invoice disputes and out-of-scope handling.
3	Freelancers and clients willing to review the prototype	Dylan (lead), Daniel (support)	Collect prototype feedback on scope guardian UX and onboarding flow. Identify usability issues. Record sessions for demo video where consent is given.
4+	Follow-ups with prior interviewees as needed	Any	Clarify outstanding questions. Validate any

			pivots made based on earlier findings.

Interview targets: freelance designers, developers, writers, consultants, and video editors in the LA area and USC network. Mix of high-volume freelancers (5+ concurrent clients) and solo practitioners.

7. Risk Mitigation

Each risk below has been assessed based on likelihood and potential impact on the project timeline or pitch. Mitigations are assigned to a primary owner and are designed to be actionable within our 6-week schedule. Where possible, we default to simpler fallbacks rather than blocking progress on any single component.

Risk	**Mitigation**	**Owner**
AI scope monitoring accuracy is low	Position as "flag for human review" - false positives are acceptable. Tune chunking, retrieval, and prompts iteratively based on real contract PDFs.	Daniel
RAG retrieval quality is poor	Use smaller, focused chunks. Test with real contract PDFs early in Week 1. Fallback: keyword-based retrieval.	Daniel
Cold start (no real users to demo)	Seed database with realistic sample freelancer and client profiles. Demo with team members role-playing both sides of a project.	Dylan
Android app scope too large	Prioritize web app. Android can be scoped down to a simplified chat-only view if timeline is at risk.	Angela
Interview recruitment is slow	Begin scheduling in Week 0 using USC network and personal connections. Backup: recruit from freelancer communities and online platforms.	Dylan
LLM API costs exceed budget	Default to Ollama for local development. Switch to OpenAI API only for demo and pitch. Monitor token usage per request.	Daniel
Matching quality is poor without enough profiles	Seed with diverse synthetic profiles across roles. Tune embedding similarity thresholds based on test results.	Dylan + Daniel